

Question:
At birth, the least developed part of the brain is the

Answer Choices:

A. visual system
B. thalamus
C. frontal lobe
D. brain stem
E. parietal lobe
F. cerebellum
G. cortex
H. basal ganglia
I. hypothalamus
J. limbic system

Reasoning + Answer:

<think>
The brain regions that mediate the most basic , immediate requirements for survival – automatic regulation of breathing and heart rate , arousal , and peristalsis – must be functional from the outset . These functions map primarily onto the brain stem (mid brain , pons , medulla). In contrast , association cortices , higher-order limbic circuits , and cerebellar circuits responsible for complex motor coordination and higher-order cognition continue to mature postnatally (e.g ., synaptogenesis , myelination , refinement of connectivity), making them less developed at birth .

Other listed structures are either largely mature at birth or are not the least developed :
- Thalamus and hypothalamus are well formed and essential for early regulation (e.g ., sleep-wake cycles , autonomic set point).
- Basal ganglia circuits have early preparatory roles (e.g ., suck/swallow , walking).
- Frontal , parietal lobes , and cortex undergo substantial postnatal development but are not the least developed at birth .

Therefore , the least developed part of the brain at birth is the cerebellum .

Correct answer : F . cerebellum
</think>
<answer>
F . cerebellum
</answer>

